2023 Summer Qual (Master)

$\boxed{1}$.

(a) ① $(AA^-A)^T = A^T(A^-)^T A^T = A^T$

$\Rightarrow (A^T)^- = (A^-)^T$

② Let x is a solution of $Ax = c$

$AA^-Ax = Ax = c$

$\Rightarrow A\underbrace{A^-c}_{=x} = c$; A^-c is a solution to $Ax = c$

③ $A\{A^-c + (I - A^-A)h\}$

$= AA^-c + Ah - \underbrace{AA^-A}_{=A}h$

$= AA^-c$

$= Ax$

$= c$; solution.

$\boxtimes$

$\boxed{2}$.

$$y^T A y \perp B y \Rightarrow \operatorname{cov}(By, y^T A y) = 0$$

$$\Rightarrow 2B\Sigma A\mu = 0$$

$$\Rightarrow BA\mu = 0 \quad \forall \mu$$

$$\Rightarrow BA = 0$$

$\boxtimes$

3.

(a) $C_X = C_{X_1} \oplus C_{X_{2\perp}}$

$H_X = H_{X_1} + H_{X_{2\perp}}$, $H_X = X(X^TX)^{-1}X^T$, $X_{2\perp} = (I - H_{X_1})X_2$

$H_{X_1} = X_1(X_1^TX_1)^{-1}X_1^T$

$\boxtimes$

(b) $\frac{1}{\sigma^2} y^T(I-H)y$

$I-H$: idempotent

$tr(I-H) = n-p-q$

$\frac{1}{2}\mu^TA\mu = 0$

$\Rightarrow \frac{1}{\sigma^2} y^T(I-H)y \sim \chi^2(n-p-q)$

$\frac{1}{\sigma^2} y^T(H-H_1)y = \frac{1}{\sigma^2} y^T H_{X_{2\perp}} y$

$H_{2\perp}^2 = H_{2\perp}$

$tr(H_{2\perp}) = tr(X_{2\perp}(X_{2\perp}^TX_{2\perp})^{-1}X_{2\perp}^T) = tr(X_{2\perp}^TX_{2\perp}(X_{2\perp}^TX_{2\perp})^{-1})$

$= tr(I_q) = q$

$\frac{1}{2}\mu^TA\mu = \frac{1}{2\sigma^2}\beta_2^TX_2^TX_2\beta_2$

$\frac{1}{\sigma^2} y^T(H-H_1)y \sim \chi^2(q, \frac{1}{2\sigma^2}\beta_2^TX_2^TX_2\beta_2)$

$\boxtimes$